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Camera optical lens including seven lenses of +−+−−+− refractive powers

Abstract

The present invention discloses a camera optical lens with seven-piece lenses including, from an object side to an image side in sequence, a first lens having a positive refractive power, a second lens having a negative refractive power, a third lens having a positive refractive power, a fourth lens having a negative refractive power, a fifth lens having a negative refractive power, a sixth lens having a positive refractive power and a seventh lens having a negative refractive power. The camera optical lens satisfies the following conditions: $-3 \leq f3/f2 \leq -2$ and $-4 \leq (R1-R2)/(R3-R4) \leq -1.5$. The camera optical lens according to the present invention has excellent optical characteristics, such as large aperture, wide-angle, and ultra-thin.

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Background/Summary

FIELD OF THE PRESENT INVENTION

(1) The present invention relates to the field of optical lens, and more particularly, to a camera optical lens suitable for handheld terminal devices, such as smart phones and digital cameras, monitors or PC lenses.

DESCRIPTION OF RELATED ART

(2) In recent years, with the rise of various smart devices, the demand for miniaturized camera optics has been increasing, and the pixel size of photosensitive devices has shrunk, coupled with the development trend of electronic products with good functions, thin and portable appearance, Therefore, miniaturized imaging optical lenses with good image quality have become the mainstream in the current market. In order to obtain better imaging quality, a multi-piece lens structure is often used. Moreover, with the development of technology and the increase of diversified needs of users, as the pixel area of the photosensitive device continues to shrink and the system's requirements for image quality continue to increase, the seven-piece lenses structure gradually appears in the lens design. There is an urgent need for a wide-angle imaging lens with excellent optical characteristics, small size, and fully corrected aberrations.

SUMMARY

(3) In the present invention, a cameral optical lens has excellent optical characteristics with large aperture stop, ultra-thin characteristic and wide-angle.

(4) According to one aspect of the present invention, a camera optical lens consisting of seven lenses includes, from an object side to an image side in sequence, a first lens having a positive refractive power, a second lens having a negative refractive power, a third lens having a positive refractive power, a fourth lens having a negative refractive power, a fifth lens having a negative refractive power, a sixth lens having a positive refractive power and a seventh lens having a negative refractive power. The camera optical lens satisfies the following conditions: $-3\leq f3/f2\leq -2$, and $-4\leq (R1-R2)/(R3-R4)\leq -1.5$. f2 denotes a focal length of the second lens, f3 denotes a focal length of the third lens, R1 denotes a central curvature radius of an object side surface of the first lens, R2 denotes a central curvature radius of an image side surface of the first lens, R3 denotes a central curvature radius of an object side surface of the second lens, and R4 denotes a central curvature radius of an image side surface of the second lens.

(5) As an improvement, the object side surface of the first lens is convex in a paraxial region and the image side surface of the first lens is concave in the paraxial region. The camera optical lens further satisfies the following conditions: $0.45\leq f1/f\leq 1.52$, $-3.37\leq (R1+R2)/(R1-R2)\leq -0.94$, and $0.07\leq d1/TTL\leq 0.21$. f denotes a focal length of the camera optical lens, f1 denotes a focal length of the first lens, d1 denotes an on-axis thickness of the first lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

(6) As an improvement, the camera optical lens further satisfies the following conditions: $0.73\leq f1/f\leq 1.21$, $-2.11\leq (R1+R2)/(R1-R2)\leq -1.17$, and $0.11\leq d1/TTL\leq 0.17$.

(7) As an improvement, the object side surface of the second lens is convex in a paraxial region and the image side surface of the second lens is concave in the paraxial region. The camera optical lens further satisfies the following conditions: $-5.68\leq f2/f\leq -1.09$, $1.02\leq (R3+R4)/(R3-R4)\leq 7.16$, and $0.02\leq d3/TTL\leq 0.07$. f denotes a focal length of the camera optical lens, d3 denotes an on-axis thickness of the second lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

(8) As an improvement, the camera optical lens further satisfies the following conditions: $-3.55\leq f2/f\leq -1.36$, $1.62\leq (R3+R4)/(R3-R4)\leq 5.73$, and $0.03\leq d3/TTL\leq 0.05$.

(9) As an improvement, the third lens has an object side surface being convex in a paraxial region and an image side surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $2.05\leq f3/f\leq 10.95$, $-16.41\leq (R5+R6)/(R5-R6)\leq -0.80$, and $0.02\leq d5/TTL\leq 0.11$. f denotes a focal length of the camera optical lens, R5 denotes a central curvature radius of the object side surface of the third lens, R6 denotes a central curvature radius of the image side surface of the third lens, d5 denotes an on-axis thickness of the third lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

(10) As an improvement, the camera optical lens further satisfies the following conditions: $3.28\leq f3/f\leq 8.76$,

$-10.26 \leq (R5+R6)/(R5-R6) \leq -1.00$, and $0.03 \leq d5/TTL \leq 0.09$.
(11) As an improvement, the fourth lens has an object side surface being concave in a paraxial region and an image side surface being convex in the paraxial region. The camera optical lens further satisfies the following conditions: $-63.50 \leq f4/f \leq -4.99$, $-30.14 \leq (R7+R8)/(R7-R8) \leq 8.52$, and $0.03 \leq d7/TTL \leq 0.16$. f denotes a focal length of the camera optical lens, f4 denotes a focal length of the fourth lens, R7 denotes a central curvature radius of the object side surface of the fourth lens, R8 denotes a central curvature radius of the image side surface of the fourth lens, d7 denotes an on-axis thickness of the fourth lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
(12) As an improvement, the camera optical lens further satisfies the following conditions: $-39.69 \leq f4/f \leq -6.24$, $-18.84 \leq (R7+R8)/(R7-R8) \leq \leq 6.82$, and $0.04 \leq d7/TTL \leq 0.13$.
(13) As an improvement, the fifth lens has an object side surface being convex in a paraxial region and an image side surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $-9.21 \leq f5/f \leq -1.20$, $0.74 \leq (R9+R10)/(R9-R10) \leq 5.52$, and $0.03 \leq d9/TTL \leq 0.12$. f denotes a focal length of the camera optical lens, f5 denotes a focal length of the fifth lens, R9 denotes a central curvature radius of the object side surface of the fifth lens, R10 denotes a central curvature radius of the image side surface of the fifth lens, d9 denotes an on-axis thickness of the fifth lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
(14) As an improvement, the camera optical lens further satisfies the following conditions: $-5.75 \leq f5/f \leq -1.50$, $1.19 \leq (R9+R10)/(R9-R10) \leq 4.41$, and $0.05 \leq d9/TTL \leq 0.09$.
(15) As an improvement, the sixth lens has an object side surface being convex in a paraxial region and an image side surface being convex in the paraxial region. The camera optical lens further satisfies the following conditions: $0.29 \leq f6/f \leq 1.19$, $-1.02 \leq (R11+R12)/(R11-R12) \leq -0.20$, and $0.05 \leq d11/TTL \leq 0.19$. f denotes a focal length of the camera optical lens, f6 denotes a focal length of the sixth lens, R11 denotes a central curvature radius of the object side surface of the sixth lens, R12 denotes a central curvature radius of the image side surface of the sixth lens, d11 denotes an on-axis thickness of the sixth lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
(16) As an improvement, the camera optical lens further satisfies the following conditions: $0.46 \leq f6/f \leq 0.95$, $-0.63 \leq (R11+R12)/(R11-R12) \leq -0.25$, and $0.08 \leq d11/TTL \leq 0.15$.
(17) As an improvement, the seventh lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $-1.38 \leq f7/f \leq -0.40$, $0.17 \leq (R13+R14)/(R13-R14) \leq 0.75$, and $0.03 \leq d13/TTL \leq 0.11$. f denotes a focal length of the camera optical lens, f7 denotes a focal length of the seventh lens, R13 denotes a central curvature radius of the object side surface of the seventh lens, R14 denotes a central curvature radius of the image side surface of the seventh lens, d13 denotes an on-axis thickness of the seventh lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
(18) As an improvement, the camera optical lens further satisfies the following conditions: $-0.86 \leq f7/f \leq -0.51$, $0.27 \leq (R13+R14)/(R13-R14) \leq 0.60$, and $0.05 \leq d13/TTL \leq 0.08$.
(19) As an improvement, an FOV of the camera optical lens is greater than or equal to 77.41°. FOV denotes a field of view of the camera optical lens in a diagonal direction.
(20) As an improvement, an FNO of the camera optical lens is less than or equal to 1.70. FNO denotes a ratio of an effective focal length of the camera optical lens to an entrance pupil diameter.
(21) As an improvement, an TTL of the camera optical lens is less than or equal to 6.49, where, TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
(22) As an improvement, the camera optical lens further satisfies the following conditions: $TTL/IH \leq 1.55$. IH denotes an image height of the camera optical lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
(23) As an improvement, the camera optical lens further satisfies the following conditions: $0.68 \leq f12/f \leq 2.41$. f denotes a focal length of the camera optical lens, and f12 denotes a combined focal length of the first lens and the second lens.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In order to explain the technical solutions in the embodiments of the present invention more clearly, the following will briefly introduce the drawings that need to be used in the description of the embodiments. Obviously, the drawings in the following description are only some embodiments of the present invention. For those of ordinary skill in the art, without creative work, other drawings can be obtained based on these drawings, among which:
(2) FIG. **1** is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 1 of the present invention;
(3) FIG. **2** is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. **1**;
(4) FIG. **3** is a schematic diagram of a lateral color of the camera optical lens shown in FIG. **1**;

(5) FIG. **4** is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. **1**;
(6) FIG. **5** is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 2 of the present invention;
(7) FIG. **6** is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. **5**;
(8) FIG. **7** is a schematic diagram of a lateral color of the camera optical lens shown in FIG. **5**;
(9) FIG. **8** is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. **5**;
(10) FIG. **9** is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 3 of the present invention;
(11) FIG. **10** is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. **9**;
(12) FIG. **11** is a schematic diagram of a lateral color of the camera optical lens shown in FIG. **9**; and
(13) FIG. **12** is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. **9**.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(14) In order to make the objects, technical solutions, and advantages of the present invention more apparent, the embodiments of the present invention will be described in detail below. However, it will be apparent to the one skilled in the art that, in the various embodiments of the present invention, a number of technical details are presented in order to provide the reader with a better understanding of the invention. However, the technical solutions claimed in the present invention can be implemented without these technical details and various changes and modifications based on the following embodiments.

Embodiment 1

(15) As referring to the accompanying drawings, the present invention provides a camera optical lens **10**. FIG. **1** shows the camera optical lens **10** according to embodiment 1 of the present invention. The camera optical lens **10** comprises seven lenses. Specifically, from an object side to an image side, the camera optical lens **10** comprises in sequence: an aperture S**1**, a first lens L**1**, a second lens L**2**, a third lens L**3**, a fourth lens L**4**, a fifth lens L**5**, a sixth lens L**6** and a seventh lens L**7**. Optical elements like optical filter GF can be arranged between the seventh lens L**7** and an image surface Si.

(16) The first lens L**1** is made of plastic material, the second lens L**2** is made of plastic material, the third lens L**3** is made of plastic material, the fourth lens L**4** is made of plastic material, the fifth lens L**5** is made of plastic material, the sixth lens L**6** is made of plastic material, and the seventh lens L**7** is made of plastic material. In other optional embodiments, each lens may also be made of other materials.

(17) In the present embodiment, a focal length of the second lens L**2** is defined as f2, a focal length of the third lens L**3** is defined as f3, the optical lens meets the following condition: $-3\leq f3/f2\leq -2$, which specifies a ratio of the focal length of the third lens L**3** to the focal length of second lens L**2**. When the condition is satisfied, the refractive power of the second lens L**2** and the third lens L**3** can be effectively distributed, and the aberration of the camera optical lens can be corrected, thereby improving the imaging quality.

(18) A central curvature radius of an object side surface of the first lens L**1** is defined as R1, a central curvature radius of an image side surface of the first lens L**1** is defined as R2, a central curvature radius of an object side surface of the second lens L**2** is defined as R3, and a central curvature radius of an image side surface of the second lens L**2** is defined as R4. The camera optical lens **10** further satisfies the following condition: $-4\leq (R1-R2)/(R3-R4)\leq -1.5$, which specifies a ratio of a difference (R1−R2) between the central curvature radius of the object side surface of the first lens L**1** and the central curvature radius of the image side surface of the first lens L**1** to a difference (R3−R4) between the central curvature radius of the object side surface of the second lens and the central curvature radius of the image side surface of the second lens L**2**. When the condition is satisfied, a sensitivity of decentering of the second lens L**2** in the camera optical lens **10** can be reduced.

(19) In the present embodiment, the object side surface of the first lens L**1** is convex in a paraxial region, the image side surface of the first lens L**1** is concave in the paraxial region, and the first lens L**1** has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the first lens L**1** can also be arranged as other concave side surface or convex side surface, such as, concave object side surface and convex image side surface and so on.

(20) A focal length of the camera optical lens **10** is defined as f, and a focal length of the first lens L**1** is defined as f1. The camera optical lens **10** further satisfies the following condition: $0.45\leq f1/f\leq 1.52$, which specifies a ratio of the focal length of the first lens L**1** to the focal length of the camera optical lens **10**. When the condition is satisfied, the first lens has an appropriate positive refractive power, which is beneficial for reducing an aberration of the camera optical lens **10** and developing ultra-thin and wide-angle lenses. Preferably, the following condition shall be satisfied, $0.73\leq f1/f\leq 1.21$.

(21) The central curvature radius of the object side surface of the first lens L**1** is defined as R1, and the central curvature radius of the image side surface of the first lens L**1** is defined as R2. The camera optical lens **10** further satisfies the following condition: $-3.37\leq (R1+R2)/(R1-R2)\leq -0.94$. This condition reasonably controls a shape of the first lens L**1**, so that the first lens L**1** can effectively correct a spherical aberration of the camera optical lens **10**. Preferably, the following condition shall be satisfied, $-2.11\leq (R1+R2)/(R1-R2)\leq -1.17$.

(22) An on-axis thickness of the first lens L**1** is defined as d1. A total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along an optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: $0.07\leq d1/TTL\leq 0.21$. When the condition is satisfied, it benefits for realizing an ultra-thin effect. Preferably, the following condition shall be satisfied, $0.11\leq d1/TTL\leq 0.17$.

(23) In the present embodiment, the object side surface of the second lens L**2** is convex in the paraxial region, the image side surface of the second lens L**2** is concave in the paraxial region, and the second lens L**2** has a negative refractive power.

In other optional embodiments, the object side surface and the image side surface of the second lens L**2** can also be arranged as other concave side surface or convex side surface, such as, concave object side surface and convex image side surface and so on.

(24) The focal length of the camera optical lens **10** is defined as f, and the focal length of the second lens L**2** is defined as f2. The camera optical lens **10** further satisfies the following condition: $-5.68 \le f2/f \le -1.09$. It is beneficial for correcting the aberration of the camera optical lens **10** by controlling the positive refractive power of the second lens L**2** being within reasonable range. Preferably, the following condition shall be satisfied, $-3.55 \le f2/f \le -1.36$.

(25) The central curvature radius of the object side surface of the second lens L**2** is defined as R3, and the central curvature radius of the image side surface of the second lens L**2** is defined as R4. The camera optical lens **10** further satisfies the following condition: $1.02 \le (R3+R4)/(R3-R4) \le 7.16$, which specifies a shape of the second lens L**2**. When the condition is satisfied, as the camera optical lens **10** develops toward the ultra-thin and wide-angle lenses, it is beneficial for correcting an on-axis chromatic aberration. Preferably, the following condition shall be satisfied, $1.62 \le (R3+R4)/(R3-R4) \le 5.73$.

(26) An on-axis thickness of the second lens L**2** is defined as d3. The total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: $0.02 \le d3/TTL \le 0.07$. When the condition is satisfied, it is beneficial for producing ultra-thin lenses. Preferably, the following condition shall be satisfied, $0.03 \le d3/TTL \le 0.05$.

(27) In the present embodiment, an object side surface of the third lens L**3** is convex in the paraxial region, an image side surface of the third lens L**3** is concave in the paraxial region, and the third lens L**3** has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the third lens L**3** can also be arranged as other concave side surface or convex side surface, such as, concave object side surface and convex image side surface and so on.

(28) The focal length of the camera optical lens **10** is defined as f, and the focal length of the third lens L**3** is defined as f3. The camera optical lens **10** further satisfies the following condition: $2.05 \le f3/f \le 10.95$. By a reasonable distribution of the refractive power, which makes it is possible that the camera optical lens **10** has an excellent imaging quality and a lower sensitivity. Preferably, the following condition shall be satisfied, $3.28 \le f3/f \le 8.76$.

(29) A central curvature radius of the object side surface of the third lens L**3** is defined as R5, and a central curvature radius of the image side surface of the third lens L**3** is defined as R6. The camera optical lens **10** further satisfies the following condition: $-16.41 \le (R5+R6)/(R5-R6) \le -0.80$, which specifies a shape of the third lens L**3**. It is beneficial for molding the third lens L**3**. When the condition is satisfied, a degree of deflection of light passing through the lens can be alleviated, and the aberration can be reduced effectively. Preferably, the following condition shall be satisfied, $-10.26 \le (R5+R6)/(R5-R6) \le -1.00$.

(30) An on-axis thickness of the third lens L**3** is defined as d5. The total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: $0.02 \le d5/TTL \le 0.11$, which benefits for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.03 \le d5/TTL \le 0.09$.

(31) In the present embodiment, an object side surface of the fourth lens L**4** is concave in the paraxial region, an image side surface of the fourth lens L**4** is convex in the paraxial region, and the fourth lens L**4** has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the fourth lens L**4** can also be arranged as other convex side surface or concave side surface, such as, convex object side surface and concave image side surface and so on.

(32) The focal length of the camera optical lens **10** is defined as f, and a focal length of the fourth lens L**4** is defined as f4. The camera optical lens **10** further satisfies the following condition: $-63.50 \le f4/f \le -4.99$. By a reasonable distribution of the refractive power, which makes it is possible that the camera optical lens **10** has the excellent imaging quality and the lower sensitivity. Preferably, the following condition shall be satisfied, $-39.69 \le f4/f \le -6.24$.

(33) A curvature radius of the object side surface of the fourth lens L**4** is defined as R7, and a central curvature radius of the image side surface of the fourth lens L**4** is defined as R8. The camera optical lens further satisfies the following condition: $-30.14 \le (R7+R8)/(R7-R8) \le 8.52$, which specifies a shape of the fourth lens L**4**. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for solving the problems, such as correcting an off-axis aberration. Preferably, the following condition shall be satisfied, $-18.84 \le (R7+R8)/(R7-R8) \le 6.82$.

(34) An on-axis thickness of the fourth lens L**4** is defined as d7. The total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: $0.03 \le d7/TTL \le 0.16$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.04 \le d7/TTL \le 0.13$.

(35) In the present embodiment, an object side surface of the fifth lens L**5** is convex in the paraxial region, an image side surface of the fifth lens L**5** is concave in the paraxial region, and the fifth lens L**5** has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the fifth lens L**5** can also be arranged as other convex side surface or concave side surface, such as, concave object side surface and convex image side surface and so on.

(36) The focal length of the camera optical lens **10** is defined as f, and a focal length of the fifth lens L**5** is defined as f5. The camera optical lens **10** further satisfies the following condition: $-9.21 \le f5/f \le -1.20$. When the condition is satisfied, a light angle of the camera optical lens **10** can be smoothed effectively and a sensitivity of the tolerance can be reduced. Preferably, the following condition shall be satisfied, $-5.75 \le f5/f \le -1.50$.

(37) A central curvature radius of the object side surface of the fifth lens L**5** is defined as R9, and a central curvature radius of the image side surface of the fifth lens L**5** is defined as R10. The camera optical lens further satisfies the following

condition: 0.74≤(R9+R10)/(R9−R10)≤5.52, which specifies a shape of the fifth lens L**5**. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, 1.19≤(R9+R10)/(R9−R10)≤4.41.
(38) An on-axis thickness of the fifth lens L**5** is defined as d9. The total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: 0.03≤d9/TTL≤0.12. When the condition is satisfied, it is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, 0.05≤d9/TTL≤0.09.
(39) In the present embodiment, an object side surface of the sixth lens L**6** is convex in the paraxial region, an image side surface of the sixth lens L**6** is convex in the paraxial region, and the sixth lens L**6** has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the sixth lens L**6** can be arranged as other convex side surface or concave side surface, such as, concave object side surface and concave image side surface and so on.
(40) The focal length of the camera optical lens **10** is defined as f, and a focal length of the sixth lens L**6** is defined as f6. The camera optical lens further satisfies the following condition: 0.29≤f6/f≤1.19. By a reasonable distribution of the refractive power, which makes it is possible that the camera optical lens **10** has the excellent imaging quality and the lower sensitivity. Preferably, the following condition shall be satisfied, 0.46≤f6/f≤0.95.
(41) A central curvature radius of the object side surface of the sixth lens L**6** is defined as R11, and a central curvature radius of the image side surface of the sixth lens L**6** is defined as R12. The camera optical lens further satisfies the following condition: −1.02≤(R11+R12)/(R11−R12)≤−0.20, which specifies a shape of the sixth lens L**6**. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it benefits for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, −0.63≤(R11+R12)/(R11−R12)≤−0.25.
(42) An on-axis thickness of the sixth lens L**6** is defined as d11. The total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens further satisfies the following condition: 0.05≤d11/TTL≤0.19, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, 0.08≤d11/TTL≤0.15.
(43) In the present embodiment, an object side surface of the seventh lens L**7** is concave in the paraxial region, an image side surface of the seventh lens L**7** is concave in the paraxial region, and the seventh lens L**7** has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the seventh lens L**7** can be arranged as other convex side surface or concave side surface, such as, convex object side surface and convex image side surface and so on.
(44) The focal length of the camera optical lens **10** is defined as f, and a focal length of the seventh lens L**7** is defined as f7. The camera optical lens **10** further satisfies the following condition: −1.38≤f7/f≤−0.40. By a reasonable distribution of the refractive power, which makes it is possible that the camera optical lens **10** has the excellent imaging quality and the lower sensitivity. Preferably, the following condition shall be satisfied, −0.86≤f7/f≤−0.51.
(45) A central curvature radius of the object side surface of the seventh lens L**7** is defined as R13, and a central curvature radius of the image side surface of the seventh lens L**7** is defined as R14. The camera optical lens **10** further satisfies the following condition: 0.17≤(R13+R14)/(R13−R14)≤0.75, which specifies a shape of the seventh lens L**7**. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, 0.27≤(R13+R14)/(R13−R14)≤0.60.
(46) An on-axis thickness of the seventh lens L**7** is defined as d13. The total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens further satisfies the following condition: 0.03≤d13/TTL≤0.11, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, 0.05≤d13/TTL≤0.08.
(47) In the present embodiment, the total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along an optical axis is defined as TTL. The TTL of the camera optical lens **10** is smaller than or equal to 6.49, thereby achieving the ultra-thin effect. Preferably, the TTL is smaller than or equal to 6.19.
(48) In the present embodiment, an image height of the camera optical lens **10** is defined as IH. The total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along an optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: TTL/IH≤1.55, thereby achieving the ultra-thin performance. Preferably, the following condition shall be satisfied, TTL/IH≤1.50.
(49) In the present embodiment, a field of view of the camera optical lens **10** in a diagonal direction is defined as FOV. The FOV is greater than or equal to 77.41°, thereby achieving the wide-angle performance. Preferably, the FOV is greater than or equal to 78.20°.
(50) In the present embodiment, the focal length of the camera optical lens **10** is f, and a combined focal length of the first lens L**1** and the second lens L**2** is defined as f12. The camera optical lens **10** further satisfies the following condition: 0.68≤f12/f≤2.41. This condition can eliminate aberration and distortion of the camera optical lens **10**, reduce a back focal length of the camera optical lens **10**, and maintain the miniaturization of the camera lens system group. Preferably, the following condition shall be satisfied, 1.08≤f12/f≤1.93.
(51) In the present embodiment, an F number (FNO) of the camera optical lens **10** is smaller than or equal to 1.70, thereby achieving a large aperture and good imaging performance. Preferably, the FNO of the camera optical lens **10** is smaller than or equal to 1.67.
(52) When the above conditions are satisfied, which makes it is possible that the camera optical lens has excellent optical

performances, and meanwhile can meet design requirements of ultra-thin, wide-angle and large aperture. According the characteristics of the camera optical lens **10**, it is particularly suitable for a mobile camera lens component and a WEB camera lens composed of high pixel CCD, CMOS.

(53) The following examples will be used to describe the camera optical lens **10** of the present invention. The symbols recorded in each example will be described as follows. The focal length, on-axis distance, central curvature radius, on-axis thickness, inflexion point position, and arrest point position are all in units of mm.

(54) TTL: the total optical length from the object side surface of the first lens L**1** to the image surface Si of the camera optical lens **10** along the optical axis, the unit of TTL is mm.

(55) F number (FNO): refers to a ratio of an effective focal length of the camera optical lens **10** to an entrance pupil diameter (ENPD).

(56) Preferably, inflexion points and/or arrest points can also be arranged on the object side surface and/or image side surface of the lens, so that the demand for high quality imaging can be satisfied, the description below can be referred for specific implementable scheme.

(57) The design information of the camera optical lens **10** in Embodiment 1 of the present invention is shown in the tables 1 and 2.

(58) TABLE-US-00001 TABLE 1 R d nd vd S1 ∞ d0 = −0.586 R1 2.070 d1 = 0.831 nd1 1.5450 v1 55.81 R2 11.970 d2 = 0.143 R3 9.987 d3 = 0.240 nd2 1.6700 v2 19.39 R4 3.398 d4 = 0.226 R5 3.668 d5 = 0.240 nd3 1.6700 v3 19.39 R6 4.686 d6 = 0.312 R7 −19.354 d7 = 0.631 nd4 1.5450 v4 55.81 R8 −25.625 d8 = 0.040 R9 8.887 d9 = 0.400 nd5 1.5661 v5 37.71 R10 5.087 d10 = 0.229 R11 2.740 d11 = 0.649 nd6 1.5450 v6 55.81 R12 −7.399 d12 = 0.629 R13 −5.398 d13 = 0.390 nd7 1.5346 v7 55.69 R14 2.638 d14 = 0.204 R15 ∞ d15 = 0.210 ndg 1.5163 vg 64.14 R16 ∞ d16 = 0.486 where, the meaning of the various symbols is as follows. S1: aperture; R: curvature radius of an optical surface, a central curvature radius for a lens; R1: central curvature radius of the object side surface of the first lens L1; R2: central curvature radius of the image side surface of the first lens L1; R3: central curvature radius of the object side surface of the second lens L2; R4: central curvature radius of the image side surface of the second lens L2; R5: central curvature radius of the object side surface of the third lens L3; R6: central curvature radius of the image side surface of the third lens L3; R7: central curvature radius of the object side surface of the fourth lens L4; R8: central curvature radius of the image side surface of the fourth lens L4; R9: central curvature radius of the object side surface of the fifth lens L5; R10: central curvature radius of the image side surface of the fifth lens L5; R11: central curvature radius of the object side surface of the sixth lens L6; R12: central curvature radius of the image side surface of the sixth lens L6; R13: central curvature radius of the object side surface of the seventh lens L7; R14: central curvature radius of the image side surface of the seventh lens L7; R15: central curvature radius of an object side surface of the optical filter GF; R16: curvature radius of an image side surface of the optical filter GF; d: on-axis thickness of a lens and an on-axis distance between lenses; d0: on-axis distance from the aperture S1 to the object side surface of the first lens L1; d1: on-axis thickness of the first lens L1; d2: on-axis distance from the image side surface of the first lens L1 to the object side surface of the second lens L2; d3: on-axis thickness of the second lens L2; d4: on-axis distance from the image side surface of the second lens L2 to the object side surface of the third lens L3; d5: on-axis thickness of the third lens L3; d6: on-axis distance from the image side surface of the third lens L3 to the object side surface of the fourth lens L4; d7: on-axis thickness of the fourth lens L4; d8: on-axis distance from the image side surface of the fourth lens L4 to the object side surface of the fifth lens L5; d9: on-axis thickness of the fifth lens L5; d10: on-axis distance from the image side surface of the fifth lens L5 to the object side surface of the sixth lens L6; d11: on-axis thickness of the sixth lens L6; d12: on-axis distance from the image side surface of the sixth lens L5 to the object side surface of the seventh lens L7; d13: on-axis thickness of the seventh lens L7; d14: on-axis distance from the image side surface of the seventh lens L7 to the object side surface of the optical filter GF; d15: on-axis thickness of the optical filter GF; d16: on-axis distance from the image side surface of the optical filter GF to the image surface; nd: refractive index of d line (d-line is green light with wavelength of 550 nm); nd1: refractive index of d line of the first lens L1; nd2: refractive index of d line of the second lens L2; nd3: refractive index of d line of the third lens L3; nd4: refractive index of d line of the fourth lens L4; nd5: refractive index of d line of the fifth lens L5; nd6: refractive index of d line of the sixth lens L6; nd7: refractive index of d line of the seventh lens L7; ndg: refractive index of d line of the optical filter GF; vd: abbe number; v1: abbe number of the first lens L1; v2: abbe number of the second lens L2; v3: abbe number of the third lens L3; v4: abbe number of the fourth lens L4; v5: abbe number of the fifth lens L5; v6: abbe number of the sixth lens L6; v7: abbe number of the seventh lens L7; vg: abbe number of the optical filter GF;

(59) Table 2 shows the aspherical surface data of the camera optical lens **10** in Embodiment 1 of the present invention.

(60) TABLE-US-00002 TABLE 2 Conic coefficient Aspherical surface coefficients k A4 A6 A8 A10 A12 R1 0.0000E+00 9.2334E−04 3.9229E−03 −1.0817E−02 1.5715E−02 −1.1739E−02 R2 0.0000E+00 −4.1746E−02 8.2400E−02 −1.0528E−01 9.4574E−02 −5.8040E−02 R3 0.0000E+00 −1.1012E−01 2.3034E−01 −2.9310E−01 2.8043E−01 −1.9628E−01 R4 0.0000E+00 −1.1793E−01 2.4359E−01 −3.4956E−01 4.2013E−01 −3.9146E−01 R5 0.0000E+00 −8.1034E−02 3.5278E−02 −6.4373E−02 7.1158E−02 −4.6994E−02 R6 00000E+00 −2.9306E−02 −4.1166E−02 1.2913E−01 −2.7765E−01 3.7553E−01 R7 0.0000E+00 −2.6203E−02 1.0492E−01 −5.2748E−01 1.2994E+00 −1.9217E+00 R8 0.0000E+00 −3.4415E−01 7.7304E−01 −1.1759E+00 1.0193E+00 −5.0155E−01 R9 0.0000E+00 −4.6192E−01 9.5647E−01 −1.3229E+00 1.1244E+00 −6.0333E−01 R10 0.0000E+00 −2.6718E−01 2.6489E−01 −2.1404E−01 1.1861E−01 −4.5436E−02 R11 −2.9884E−03 −6.5863E−02 −5.2286E−03 2.2758E−02 −2.0841E−02 7.3346E−03 R12 0.0000E+00 1.0708E−01 −9.9611E−02 6.3181E−02 −2.8349E−02 8.0205E−03 R13 0.0000E+00 −8.5073E−02 −2.6709E−02 3.5151E−02 −1.1517E−02 1.9448E−03 R14 −4.9284E−01 −1.2358E−01 3.1019E−02

−4.3110E−03 −2.0140E−04 2.2938E−04 Conic coefficient Aspherical surface coefficients k A14 A16 A18 A20 R1 0.0000E+00 4.4312E−03 −7.0286E−04 0.0000E+00 0.0000E+00 R2 0.0000E+00 2.2033E−02 −4.5529E−03 3.7383E−04 0.0000E+00 R3 0.0000E+00 9.2917E−02 −2.5668E−02 3.1085E−03 0.0000E+00 R4 0.0000E+00 2.5056E−01 −9.4364E−02 1.5964E−02 0.0000E+00 R5 0.0000E+00 2.7507E−03 1.3038E−02 −4.3332E−03 0.0000E+00 R6 0.0000E+00 −3.1091E−01 1.4868E−01 −3.5317E−02 2.8630E−03 R7 00000E+00 1.7521E+00 −9.6483E−01 2.9294E−01 −3.7372E−02 R8 0.0000E+00 1.1697E−01 3.9120E−03 −8.2303E−03 1.3071E−03 R9 0.0000E+00 2.0850E−01 −4.5257E−02 5.6094E−03 −3.0168E−04 R10 0.0000E+00 1.2821E−02 −2.5356E−03 2.9747E−04 −1.4954E−05 R11 −2.9884E−03 −6.4718E−04 −3.8859E−04 1.3267E−04 −1.2455E−05 R12 0.0000E+00 −1.3662E−03 1.3468E−04 −6.9809E−06 1.4357E−07 R13 0.0000E+00 −1.8897E−04 1.0430E−05 −2.9076E−07 2.8255E−09 R14 −4.9284E−01 −4.8539E−05 5.1818E−06 −2.8361E−07 6.2839E−09

(61) For convenience, an aspheric surface of each lens surface uses the aspheric surfaces shown in the below condition (1). However, the present invention is not limited to the aspherical polynomials form shown in the condition (1).

(62)

$$z = (cr^2)/\{1+[1-(k+1)(c^2r^2)]^{1/2}\} + A4r^4 + A6r^6 + A8r^8 + A10r^{10} + A12r^{12} + A14r^{14} + A16r^{16} + A18r^{18} + A20r^{20} \quad (1)$$

(63) Where, K is a conic coefficient, A4, A6, A8, A10, A12, A14, A16, A18, A20 are aspheric surface coefficients. c is the curvature at the center of the optical surface. r is a vertical distance between a point on an aspherical curve and the optic axis, and z is an aspherical depth (a vertical distance between a point on an aspherical surface, having a distance of r from the optic axis, and a surface tangent to a vertex of the aspherical surface on the optic axis).

(64) Table 3 and Table 4 show design data of inflexion points and arrest points of respective lens in the camera optical lens **10** according to Embodiment 1 of the present invention. P1R1 and P1R2 represent the object side surface and the image side surface of the first lens L**1**, P2R1 and P2R2 represent the object side surface and the image side surface of the second lens L**2**, P3R1 and P3R2 represent the object side surface and the image side surface of the third lens L**3**, P4R1 and P4R2 represent the object side surface and the image side surface of the fourth lens L**4**, P5R1 and P5R2 represent the object side surface and the image side surface of the fifth lens L**5**, P6R1 and P6R2 represent the object side surface and the image side surface of the sixth lens L**6**, and P7R1 and P7R2 represent the object side surface and the image side surface of the seventh lens L**7**. The data in the column named "inflexion point position" refers to vertical distances from inflexion points arranged on each lens surface to the optical axis of the camera optical lens **10**. The data in the column named "arrest point position" refers to vertical distances from arrest points arranged on each lens surface to the optical axis of the camera optical lens **10**.

(65) TABLE-US-00003 TABLE 3 Number of Inflexion Inflexion Inflexion Inflexion inflexion point point point point points position 1 position 2 position 3 position 4 P1R1 1 1.425 / / / P1R2 1 1.085 / / / P2R1 0 / / / / P2R2 0 / / / / P3R1 1 0.585 / / / P3R2 3 0.685 1.075 1.315 / P4R1 1 1.235 / / / P4R2 1 1.435 / / / P5R1 2 0.155 1.195 / / P5R2 2 0.275 1.335 / / P6R1 3 0.765 1.805 1.995 / P6R2 4 0.395 0.975 1.995 2.245 P7R1 3 0.415 2.755 2.955 / P7R2 4 0.575 2.725 2.975 3.285

(66) TABLE-US-00004 TABLE 4 Number of Arrest point Arrest point Arrest point arrest points position 1 position 2 position 3 P1R1 0 / / / P1R2 1 1.375 / / P2R1 0 / / / P2R2 0 / / / P3R1 1 0.955 / / P3R2 0 / / / P4R1 0 / / / P4R2 0 / / / P5R1 2 0.275 1.685 / P5R2 2 0.515 1.695 / P6R1 1 1.235 / / P6R2 0 / / / P7R1 1 2.435 / / P7R2 1 1.135 / /

(67) FIG. **2** and FIG. **3** respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 650 nm, 555 nm and 470 nm after passing the camera optical lens **10** according to Embodiment 1. FIG. **4** illustrates a field curvature and a distortion of light with a wavelength of 555 nm after passing the camera optical lens **10** according to Embodiment 1, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

(68) Table 13 shows various values of Embodiments 1, 2 and 3 and values corresponding to parameters which are specified in the above conditions.

(69) As shown in Table 13, Embodiment 1 satisfies the above conditions.

(70) In the present embodiment, the entrance pupil diameter (ENPD) of the camera optical lens **10** is 2.916 mm. The image height of 1.0H is 4.000 mm. The FOV is 79.00°. Thus, the camera optical lens **10** satisfies design requirements of large aperture, ultra-thin and wide-angle while the on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

Embodiment 2

(71) Embodiment 2 is basically the same as Embodiment 1, the meaning of its symbols is the same as that of Embodiment 1, in the following, only the differences are listed.

(72) FIG. **5** shows a schematic diagram of a structure of a camera optical lens **20** according to Embodiment 2 of the present invention. Table 5 and table 6 show the design data of a camera optical lens **20** in Embodiment 2 of the present invention.

(73) TABLE-US-00005 TABLE 5 R d nd vd S1 ∞ d0 = −0.560 R1 2.003 d1 = 0.838 nd1 1.5450 v1 55.81 R2 7.839 d2 = 0.089 R3 4.266 d3 = 0.260 nd2 1.6700 v2 19.39 R4 2.788 d4 = 0.359 R5 8.057 d5 = 0.240 nd3 1.6700 v3 19.39 R6 14.627 d6 = 0.229 R7 −8.413 d7 = 0.594 nd4 1.5450 v4 55.81 R8 −9.609 d8 = 0.040 R9 8.076 d9 = 0.400 nd5 1.5661 v5 37.71 R10 3.514 d10 = 0.198 R11 2.823 d11 = 0.747 nd6 1.5450 v6 55.81 R12 −5.213 d12 = 0.560 R13 −6.771 d13 = 0.390 nd7 1.5346 v7 55.69 R14 2.343 d14 = 0.204 R15 ∞ d15 = 0.210 ndg 1.5163 vg 64.14 R16 ∞ d16 = 0.503

(74) Table 6 shows aspherical surface data of each lens of the camera optical lens **20** in Embodiment 2 of the present invention.

(75) TABLE-US-00006 TABLE 6 Conic coefficient Aspherical surface coefficients k A4 A6 A8 A10 A12 R1 0.0000E+00 5.8559E−03 −1.7380E−02 3.9278E−02 −5.1339E−02 4.1459E−02 R2 0.0000E+00 −1.3395E−01 2.4683E−01

−3.2095E−01 3.0853E−01 −2.1228E−01 R3 0.0000E+00 −2.1817E−01 3.5005E−01 −3.8492E−01 3.2970E−01 −2.1684E−01 R4 0.0000E+00 −1.2508E−01 1.4415E−01 4.4522E−03 −2.4493E−01 3.7697E−01 R5 0.0000E+00 −5.4304E−02 −5.9500E−02 2.5040E−01 −6.2495E−01 9.7575E−01 R6 0.0000E+00 −2.2362E−02 −9.5462E−02 2.8226E−01 −4.9868E−01 5.4753E−01 R7 0.0000E+00 3.6149E−03 2.3864E−02 −3.2012E−01 9.4654E−01 −1.5103E+00 R8 0.0000E+00 −1.4511E−01 2.2514E−01 −2.9965E−01 8.6301E−02 1.9936E−01 R9 0.0000E+00 −2.7312E−01 3.8568E−01 −4.0786E−01 1.3203E−01 1.4833E−01 R10 0.0000E+00 −2.4984E−01 2.0113E−01 −1.3824E−01 5.8973E−02 −1.4009E−02 R11 3.6420E−01 −6.3181E−02 −2.7356E−02 5.9232E−02 −5.6285E−02 3.0825E−02 R12 0.0000E+00 1.0691E−01 −1.0238E−01 6.3412E−02 −2.5964E−02 6.4188E−03 R13 0.0000E+00 −8.5890E−02 −3.4738E−02 3.7500E−02 −1.1431E−02 1.7935E−03 R14 −5.7954E−01 −1.3951E−01 4.0103E−02 −8.5018E−03 1.0804E−03 −1.7781E−05 Conic coefficient Aspherical surface coefficients k A14 A16 A18 A20 R1 0.0000E+00 −2.0436E−02 5.6363E−03 −6.8727E−04 0.0000E+00 R2 0.0000E+00 9.7790E−02 −2.7885E−02 4.2982E−03 −2.6083E−04 R3 0.0000E+00 1.0086E−01 −2.8159E−02 3.4907E−03 0.0000E−00 R4 0.0000E+00 −2.9543E−01 1.3412E−01 −3.4789E−02 4.6865E−03 R5 0.0000E+00 −9.7070E−01 5.8472E−01 −1.9167E−01 2.6295E−02 R6 0.0000E+00 −3.5539E−01 1.1387E−01 −5.7477E−03 −3.5447E−03 R7 0.0000E+00 1.4467E+00 −8.3125E−01 2.6236E−01 −3.4674E−02 R8 0.0000E+00 −2.5073E−01 1.3099E−01 −3.4194E−02 3.6987E−03 R9 0.0000E+00 −1.7897E−01 8.2153E−02 −1.8223E−02 1.6177E−03 R10 0.0000E+00 2.2933E−03 −4.4553E−04 7.7796E−05 −5.7449E−06 R11 3.6420E−01 −1.0893E−02 2.4063E−03 −2.9363E−04 1.4864E−05 R12 0.0000E+00 −8.9348E−04 6.0719E−05 −9.7185E−07 −5.6747E−08 R13 0.0000E+00 −1.5766E−04 7.3421E−06 −1.3648E−07 −3.2008E−10 R14 −5.7954E−01 −1.8212E−05 2.8342E−06 −1.7789E−07 4.1733E−09

(76) Table 7 and table 8 show design data of inflexion points and arrest points of respective lens in the camera optical lens **20** according to Embodiment 2 of the present invention.

(77) TABLE-US-00007 TABLE 7 Number of Inflexion Inflexion Inflexion Inflexion inflexion point point point point points position 1 position 2 position 3 position 4 P1R1 1 1.395 / / / P1R2 3 0.415 0.595 1.055 / P2R1 2 0.465 0.545 / / P2R2 0 / / / / P3R1 2 0.415 1.145 / / P3R2 2 0.395 1.105 / / P4R1 1 1.225 / / / P4R2 1 1.415 / / / P5R1 2 0.215 1.395 / / P5R2 2 0.355 1.355 / / P6R1 2 0.735 1.815 / / P6R2 4 0.615 0.695 2.015 2.205 P7R1 3 1.445 2.825 2.925 / P7R2 4 0.585 2.775 3.045 3.315

(78) TABLE-US-00008 TABLE 8 Number of Arrest point Arrest point Arrest point arrest points position 1 position 2 position 3 P1R1 0 / / / P1R2 1 1.305 / / P2R1 0 / / / P2R2 0 / / / P3R1 1 0.705 / / P3R2 2 0.675 1.235 / P4R1 0 / / / P4R2 0 / / / P5R1 1 0.395 / / P5R2 2 0.685 1.755 / P6R1 1 1.255 / / P6R2 0 / / / P7R1 1 2.445 / / P7R2 1 1.165 / /

(79) FIG. **6** and FIG. **7** respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 650 nm, 555 nm and 470 nm after passing the camera optical lens **20** according to Embodiment 2. FIG. **8** illustrates a field curvature and a distortion of light with a wavelength of 555 nm after passing the camera optical lens **10** according to Embodiment 2, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

(80) As shown in Table 13, Embodiment 2 satisfies the above conditions.

(81) In the present embodiment, an entrance pupil diameter (ENPD) of the camera optical lens is 2.884 mm. An image height of 1.0H is 4.000 mm. An FOV is 78.99°. Thus, the camera optical lens **20** satisfies design requirements of large aperture, ultra-thin and wide-angle while the on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

Embodiment 3

(82) Embodiment 3 is basically the same as Embodiment 1 and involves symbols having the same meanings as Embodiment 1, and only differences therebetween will be described in the following.

(83) FIG. **9** shows a schematic diagram of a structure of a camera optical lens **30** according to Embodiment 3 of the present invention.

(84) Tables 9 and 10 show design data of a camera optical lens **30** in Embodiment 3 of the present invention.

(85) TABLE-US-00009 TABLE 9 R d nd vd S1 ∞ d0 = −0.544 R1 2.003 d1 = 0.820 nd1 1.5450 v1 55.81 R2 11.714 d2 = 0.115 R3 6.436 d3 = 0.240 nd2 1.6700 v2 19.39 R4 2.840 d4 = 0.214 R5 4.158 d5 = 0.240 nd3 1.6700 v3 19.39 R6 5.949 d6 = 0.356 R7 −23.698 d7 = 0.613 nd4 1.5450 v4 55.81 R8 −33.633 d8 = 0.080 R9 11.093 d9 = 0.400 nd5 1.5661 v5 37.71 R10 3.602 d10 = 0.158 R11 2.366 d11 = 0.755 nd6 1.5450 v6 55.81 R12 −5.691 d12 = 0.568 R13 −6.944 d13 = 0.390 nd7 1.5346 v7 55.69 R14 2.308 d14 = 0.204 R15 ∞ d15 = 0.210 ndg 1.5163 vg 64.14 R16 ∞ d16 = 0.496

(86) Table 10 shows aspherical surface data of each lens of the camera optical lens **30** in Embodiment 3 of the present invention.

(87) TABLE-US-00010 TABLE 10 Conic coefficient Aspherical surface coefficients k A4 A6 A8 A10 A12 R1 0.0000E+00 1.9172E−03 1.8079E−03 −7.9391E−03 1.4132E−02 −1.1943E−02 R2 0.0000E+00 −7.1899E−02 1.5791E−01 −2.2646E−01 2.2679E−01 −1.5480E−01 R3 0.0000E+00 −1.6785E−01 3.4332E−01 −4.6004E−01 4.5479E−01 −3.2182E−01 R4 0.0000E+00 −1.4707E−01 2.6339E−01 −2.9129E−01 2.1663E−01 −1.1335E−01 R5 0.0000E+00 −6.7822E−02 6.3356E−02 −1.5876E−01 2.6493E−01 −2.9367E−01 R6 00000E+00 −2.2439E−02 −1.3877E−02 5.3822E−02 −1.4243E−01 2.0893E−01 R7 0.0000E+00 −3.1599E−02 5.2292E−02 −3.3642E−01 9.0789E−01 −1.4417E+00 R8 0.0000E+00 −1.6966E−01 1.5551E−01 −6.0039E−02 −2.6971E−01 4.9855E−01 R9 0.0000E+00 −2.7656E−01 3.4258E−01 −2.3836E−01 −7.8481E−02 2.6957E−01 R10 00000E+00 −2.9543E−01 2.4236E−01 −1.2320E−01 1.3223E−02 2.1615E−02 R11 −1.8140E−01 −1.0547E−01 −4.8439E−03 6.8175E−02 −7.8781E−02 4.7409E−02 R12 0.0000E+00 1.1176E−01 −1.1499E−01 7.3970E−02 −3.0881E−02 7.7613E−03 R13 0.0000E+00

−1.0654E−01 −2.0435E−02 3.4167E−02 −1.1505E−02 1.9907E−03 R14 −7.6759E−01 −1.5429E−01 5.3546E−02 −1.4697E−02 2.9924E−03 −4.0718E−04 Conic coefficient Aspherical surface coefficients k A14 A16 A18 A20 R1 0.0000E+00 4.9245E−03 −8.4785E−04 0.0000E+00 0.0000E+00 R2 0.0000E+00 6.6778E−02 −1.6271E−02 1.6870E−03 0.0000E+00 R3 0 0000E+00 1.5242E−01 −4.2279E−02 5.1683E−03 0.0000E+00 R4 0.0000E+00 5.4088E−02 −2.4848E−02 6.6719E−03 0.0000E+00 R5 0.0000E+00 1.9049E−01 −6.4978E−02 9.4098E−03 0.0000E+00 R6 0.0000E+00 −1.7277E−01 7.3933E−02 −1.0441E−02 −1.0574E−03 R7 0.0000E+00 1.3925E+00 −8.0457E−01 2.5472E−01 −3.3728E−02 R8 0.0000E+00 −4.0701E−01 1.8328E−01 −4.4728E−02 4.6719E−03 R9 0.0000E+00 −2.0480E−01 7.7226E−02 −1.4965E−0.2 1.1922E−03 R10 0.0000E+00 −1.2874E−02 3.3098E−03 −4.2549E−04 2.2392E−05 R11 −1.8140E−01 −1.7691E−02 4.0210E−03 −4.9851E−04 2.5547E−05 R12 0.0000E+00 −1.1109E−03 8.1135E−05 −1.9883E−06 −3.6436E−08 R13 0.0000E+00 −2.0162E−04 1.2021E−05 −3.8882E−07 5.2148E−09 R14 −7.6759E−01 3.2707E−05 −1.2364E−06 2.3294E−09 8.0118E−10

(88) Table 11 and table 12 show Embodiment 3 design data of inflexion points and arrest points of respective lens in the camera optical lens **30** according to Embodiment 3 of the present invention.

(89) TABLE-US-00011 TABLE 11 Number of Inflexion Inflexion Inflexion Inflexion inflexion point point point point points position 1 position 2 position 3 position 4 P1R1 1 1.385 / / / P1R2 1 1.045 / / / P2R1 0 / / / / P2R2 0 / / / / P3R1 2 0.625 1.145 / / P3R2 2 0.725 1.015 / / P4R1 1 1.215 / / / P4R2 1 1.425 / / / P5R1 2 0.175 1.595 / / P5R2 2 0.315 1.355 / / P6R1 2 0.695 1.795 / / P6R2 4 0.525 0.725 1.985 2.255 P7R1 1 1.425 / / / P7R2 4 0.565 2.715 3.115 3.305

(90) TABLE-US-00012 TABLE 12 Number of Arrest point Arrest point Arrest point arrest points position 1 position 2 poition 3 P1R1 0 / / / P1R2 1 1.305 / / P2R1 0 / / / P2R2 0 / / / P3R1 1 1.015 / / P3R2 0 / / / P4R1 0 / / / P4R2 0 / / / P5R1 1 0.315 / / P5R2 2 0.605 1.785 / P6R1 1 1.255 / / P6R2 0 / / / P7R1 1 2.385 / / P7R2 1 1.135 / /

(91) FIG. **10** and FIG. **11** respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 650 nm, 555 nm and 470 nm after passing the camera optical lens **30** according to Embodiment 3. FIG. **12** illustrates a field curvature and a distortion of light with a wavelength of 555 nm after passing the camera optical lens **30** according to Embodiment 3, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

(92) Table 13 in the following lists values corresponding to the respective conditions. In the present Embodiment 3 in order to satisfy the above conditions.

(93) In the present embodiment, an entrance pupil diameter (ENPD) of the camera optical lens is 2.867 mm. An image height of 1.0H is 4.000 mm. An FOV is 79.00°. Thus, the camera optical lens **30** satisfies design requirements of large aperture, ultra-thin and wide-angle while the on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

(94) TABLE-US-00013 TABLE 13 Parameters and conditions Embodiment 1 Embodiment 2 Embodiment 3 f3/f2 −2.95 −2.04 −2.51 (R1 − R2)/(R3 − R4) −1.51 −3.95 −2.70 f 4.724 4.730 4.730 f1 4.317 4.682 4.292 f2 −7.728 −12.815 −7.722 f3 22.799 −26.142 19.383 f4 −149.999 −149.999 −149.999 f5 −21.744 −11.795 −9.564 f6 3.742 −3.463 3.161 f7 −3.249 −3.197 −3.183 f12 7.579 6.386 7.483 FNO 1.620 1.640 1.650 TTL 5.860 5.860 5.860 IH 4.000 4.000 4.000 FOV 79.00° 78.99° 79.00°

(95) It is to be understood, however, that even though numerous characteristics and advantages of the present exemplary embodiments have been set forth in the foregoing description, together with details of the structures and functions of the embodiments, the disclosure is illustrative only, and changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms where the appended claims are expressed.

Claims

1. A camera optical lens consisting of seven-piece lenses, comprising, from an object side to an image side in sequence: a first lens having a positive refractive power, a second lens having a negative refractive power, a third lens having a positive refractive power, a fourth lens having a negative refractive power, a fifth lens having a negative refractive power, a sixth lens having a positive refractive power and a seventh lens having a negative refractive power; wherein the camera optical lens satisfies the following conditions: $-3 \leqslant f3/f2 \leqslant -2$; $-4 \leqslant (R1-R2)/(R3-R4) \leqslant -1.5$; $-39.69 \leqslant f4/f \leqslant -31.71$; $-18.84 \leqslant (R7+R8)/(R7-R8) \leqslant 6.82$; and $0.04 \leqslant d7/\mathrm{TTL} \leqslant 0.13$. where, f2: a focal length of the second lens; f3: a focal length of the third lens; f4: a focal length of the fourth lens; f: a focal length of the camera optical lens; R1: a central curvature radius of an object side surface of the first lens; R2: a central curvature radius of an image side surface of the first lens; R3: a central curvature radius of an object side surface of the second lens; R4: a central curvature radius of an image side surface of the second lens, R7: a central curvature radius of the object side surface of the fourth lens; R8: a central curvature radius of the image side surface of the fourth lens; d7: an on-axis thickness of the fourth lens; and TTL: a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

2. The camera optical lens according to claim 1, wherein, the object side surface of the first lens is convex in a paraxial region and the image side surface of the first lens is concave in the paraxial region; the camera optical lens further satisfies the following conditions: $0.45 \leqslant f1/f \leqslant 1.52$; $-3.37 \leqslant (R1+R2)/(R1-R2) \leqslant -0.94$; and $0.07 \leqslant d1/\mathrm{TTL} \leqslant 0.21$; where, f: a focal length of the camera optical lens; f1: a focal length of the first lens; d1: an on-axis thickness of the first

lens; and TTL: a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
3. The camera optical lens according to claim 2 further satisfying the following conditions:
$0.73 \leqslant f1 / f \leqslant 1.21; -2.11 \leqslant (R1 + R2) / (R1 - R2) \leqslant -1.17; \text{and} 0.11 \leqslant d1 / \mathrm{TTL} \leqslant 0.17$.
4. The camera optical lens according to claim 1, wherein, the object side surface of the second lens is convex in a paraxial region and the image side surface of the second lens is concave in the paraxial region; the camera optical lens further satisfies the following conditions:
$-5.68 \leqslant f2 / f \leqslant -1.09; 1.02 \leqslant (R3 + R4) / (R3 - R4) \leqslant 7.16; \text{and} 0.02 \leqslant d3 / \mathrm{TTL} \leqslant 0.07$; where, f: a focal length of the camera optical lens; d3: an on-axis thickness of the second lens; and TTL: a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
5. The camera optical lens according to claim 4 further satisfying the following conditions:
$-3.55 \leqslant f2 / f \leqslant -1.36; 1.62 \leqslant (R3 + R4) / (R3 - R4) \leqslant 5.73; \text{and} 0.03 \leqslant d3 / \mathrm{TTL} \leqslant 0.5$.
6. The camera optical lens according to claim 1, wherein, the third lens has an object side surface being convex in a paraxial region and an image side surface being concave in the paraxial region; the camera optical lens further satisfies the following conditions: $2.05 \leqslant f3 / f \leqslant 10.95; -16.41 \leqslant (R5 + R6) / (R5 - R6) \leqslant -0.8; \text{and} 0.02 \leqslant d5 / \mathrm{TTL} \leqslant 0.11$; where, f: a focal length of the camera optical lens; R5: a central curvature radius of the object side surface of the third lens; R6: a central curvature radius of the image side surface of the third lens; d5: an on-axis thickness of the third lens; and TTL: a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
7. The camera optical lens according to claim 6 further satisfying the following conditions:
$3.28 \leqslant f3 / f \leqslant 8.76; -10.26 \leqslant (R5 + R6) / (R5 - R6) \leqslant -1.; \text{and} 0.03 \leqslant d5 / \mathrm{TTL} \leqslant 0.9$.
8. The camera optical lens according to claim 1, wherein, the fifth lens has an object side surface being convex in a paraxial region and an image side surface being concave in the paraxial region; the camera optical lens further satisfies the following conditions: $-9.21 \leqslant f5 / f \leqslant -1.2; 0.74 \leqslant (R9 + R10) / (R9 - R10) \leqslant 5.52; \text{and} 0.03 \leqslant d9 / \mathrm{TTL} \leqslant 0.12$; where, f5: a focal length of the fifth lens; R9: a central curvature radius of the object side surface of the fifth lens; R10: a central curvature radius of the image side surface of the fifth lens; and d9: an on-axis thickness of the fifth lens.
9. The camera optical lens according to claim 8 further satisfying the following conditions:
$-5.75 \leqslant f5 / f \leqslant -1.5; 1.19 \leqslant (R9 + R10) / (R9 - R10) \leqslant 4.41; \text{and} 0.05 \leqslant d9 / \mathrm{TTL} \leqslant 0.9$.
10. The camera optical lens according to claim 1, wherein, the sixth lens has an object side surface being convex in a paraxial region and an image side surface being convex in the paraxial region; the camera optical lens further satisfies the following conditions: $0.29 \leqslant f6 / f \leqslant 1.19; -1.02 \leqslant (R11 + R12) / (R11 - R12) \leqslant -0.2; \text{and} 0.05 \leqslant d11 / \mathrm{TTL} \leqslant 0.19$; where, f6: a focal length of the sixth lens; R11: a central curvature radius of the object side surface of the sixth lens; R12: a central curvature radius of the image side surface of the sixth lens; and d11: an on-axis thickness of the sixth lens.
11. The camera optical lens according to claim 10 further satisfying the following conditions:
$0.46 \leqslant f6 / f \leqslant 0.95; -0.63 \leqslant (R11 + R12) / (R11 - R12) \leqslant -0.25; \text{and} 0.08 \leqslant d11 / \mathrm{TTL} \leqslant 0.15$.
12. The camera optical lens according to claim 1, wherein, the seventh lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region; the camera optical lens further satisfies the following conditions: $-1.38 \leqslant f7 / f \leqslant -0.4; 0.17 \leqslant (R13 + R14) / (R13 - R14) \leqslant 0.75; \text{and} 0.03 \leqslant d13 / T\mathrm{TL} \leqslant 0.11$; where, f7: a focal length of the seventh lens; R13: a central curvature radius of the object side surface of the seventh lens; R14: a central curvature radius of the image side surface of the seventh lens; and d13: an on-axis thickness of the seventh lens.
13. The camera optical lens according to claim 12 further satisfying the following conditions:
$-0.86 \leqslant f7 / f \leqslant -0.51; 0.27 \leqslant (R13 + R14) / (R13 - R14) \leqslant 0.6; \text{and} 0.05 \leqslant d13 / T\mathrm{TL} \leqslant 0.8$.
14. The camera optical lens according to claim 1, wherein an FOV of the camera optical lens is greater than or equal to 77.41°, where, FOV: a field of view of the camera optical lens in a diagonal direction.
15. The camera optical lens according to claim 1, wherein an FNO of the camera optical lens is less than or equal to 1.70, where, FNO: a ratio of an effective focal length of the camera optical lens to an entrance pupil diameter.
16. The camera optical lens according to claim 1, wherein an TTL of the camera optical lens is less than or equal to 6.49.
17. The camera optical lens according to claim 1 further satisfying the following condition: TTL/IH≤1.55; where, IH: an image height of the camera optical lens.
18. The camera optical lens according to claim 1 further satisfying the following condition: 0.68≤f12/f≤2.41; where, f: a focal length of the camera optical lens; and f12: a combined focal length of the first lens and the second lens.